

WAB Exposé

Provadis School of International Management and Technology

**Performance of Various Python Runtimes for
a Bioinformatics Algorithm**

Testing the Smith-Waterman algorithm across CPython, PyPy,
and Python 3.14's native JIT

Rubin Chempananickal James
rubin.chempananickal-james@stud-provadis-hochschule.de
Matriculation Number: D876

Department: Information Technology
Module: Fortgeschrittene Programmierung
Reviewer: Prof. Dr. Jörg Daubert

03.05.2026

Contents

Exposé	1
1. Introduction	1
2. Objectives	1
3. Methodology	1
4. Planned Structure	1
References	i

Exposé

1. Introduction

2. Objectives

3. Methodology

4. Planned Structure

The paper will likely be structured as follows:

- Introduction
- Background and Related Work
- Methodology
- Results
- Discussion
- Conclusion and Future Work
- References
- AI Declaration
- Declaration of Authorship

References

- [1] T. F. Smith and M. S. Waterman, “Identification of common molecular subsequences,” *Journal of Molecular Biology*, vol. 147, no. 1, pp. 195–197, 1981, doi: 10.1016/0022-2836(81)90087-5.
- [2] S. Behnel, R. Bradshaw, C. Citro, L. Dalcin, D. S. Seljebotn, and K. Smith, “Cython: The Best of Both Worlds,” *Computing in Science & Engineering*, vol. 13, no. 2, pp. 31–39, 2011, doi: 10.1109/MCSE.2010.118.
- [3] B. Bucher and S. Ostrowski, “PEP 744 – JIT Compilation.” Accessed: Mar. 10, 2026. [Online]. Available: <https://peps.python.org/pep-0744/>
- [4] Python Software Foundation, “What's New In Python 3.14.” Accessed: Mar. 10, 2026. [Online]. Available: <https://docs.python.org/3/whatsnew/3.14.html>
- [5] The PyPy Team, “PyPy: Goals and Architecture Overview.” Accessed: Apr. 27, 2026. [Online]. Available: <https://doc.pypy.org/architecture.html#pypy-python-interpreter>
- [6] The PyPy Team, “PyPy: A Fast, Compliant Alternative Implementation of Python.” Accessed: Apr. 27, 2026. [Online]. Available: <https://www.pypy.org/>
- [7] The PyPy Team, “The RPython Language.” Accessed: Apr. 27, 2026. [Online]. Available: <https://rpython.readthedocs.io/en/latest/rpython.html>
- [8] The PyPy Team, “PyPy Speed Center.” Accessed: Apr. 27, 2026. [Online]. Available: <https://speed.pypy.org/>
- [9] G. Van Rossum and F. L. Drake, *Python 3 Reference Manual*. Scotts Valley, CA: CreateSpace, 2009.
- [10] D. Mariano, M. Ferreira, B. L. Sousa, L. H. Santos, and R. C. de Melo-Minardi, “A Brief History of Bioinformatics Told by Data Visualization,” in *Advances in Bioinformatics and Computational Biology*, J. C. Setubal and W. M. Silva, Eds., Cham: Springer International Publishing, 2020, pp. 235–246. doi: 10.1007/978-3-030-65775-8_22.

- [11] M. Gorelick and I. Ozsvald, *High performance python*. O'Reilly Media Inc., 2025.
- [12] M. Foord and C. Muirhead, *IronPython in action*. Manning Publications Co., 2009.
- [13] “Mojo: Powerful CPU + GPU Programming.” Accessed: Apr. 27, 2026. [Online]. Available: <https://modular.com/mojo>
- [14] J. Zhong, M. Hort, and F. Sarro, “Py2Cy: a genetic improvement tool to speed up python,” in *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, in GECCO '22. Boston, Massachusetts: Association for Computing Machinery, 2022, pp. 1950–1955. doi: 10.1145/3520304.3534037.
- [15] J. Akeret, L. Gamper, A. Amara, and A. Refregier, “HOPE: A Python just-in-time compiler for astrophysical computations,” *Astronomy and Computing*, vol. 10, pp. 1–8, 2015, doi: <https://doi.org/10.1016/j.ascom.2014.12.001>.
- [16] S. K. Lam, A. Pitrou, and S. Seibert, “Numba: a LLVM-based Python JIT compiler,” in *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, in LLVM '15. Austin, Texas: Association for Computing Machinery, 2015. doi: 10.1145/2833157.2833162.
- [17] S. Guelton, P. Brunet, M. Amini, A. Merlini, X. Corbillon, and A. Raynaud, “Pythran: enabling static optimization of scientific Python programs,” *Computational Science & Discovery*, vol. 8, no. 1, p. 14001, Mar. 2015, doi: 10.1088/1749-4680/8/1/014001.
- [18] A. Roghult, “Benchmarking Python Interpreters: Measuring Performance of CPython, Cython, Jython and PyPy.” 2016.
- [19] K. Jin, “Reflections on 2 years of CPython’s JIT Compiler: The good, the bad, the ugly.” Accessed: Apr. 28, 2026. [Online]. Available: <https://fidget-spinner.github.io/posts/jit-reflections.html>
- [20] B. Hernández-Bermejo, V. Fairén, and A. Sorribas, “Power-law modeling based on least-squares criteria: consequences for system analysis and simulation,” *Mathematical biosciences*, vol. 167, no. 2, pp. 87–107, 2000.

- [21] Intel Corporation, “Intel® Core™ i5-10210U Processor (6M Cache, up to 4.20 GHz) Specifications.” Accessed: Feb. 26, 2026. [Online]. Available: <https://www.intel.com/content/www/us/en/products/sku/195436/intel-core-i510210u-processor-6m-cache-up-to-4-20-ghz/specifications.html>
- [22] T. J. Wheeler and S. R. Eddy, “nhmmer: DNA homology search with profile HMMs,” *Bioinformatics*, vol. 29, no. 19, pp. 2487–2489, 2013, doi: 10.1093/bioinformatics/btt403.
- [23] J. W. Jurka, “Chapter Thirty-Nine - Approaches to Identification and Analysis of Interspersed Repetitive DNA Sequences,” *Automated DNA Sequencing and Analysis*. Academic Press, San Diego, pp. 294–298, 1994. doi: <https://doi.org/10.1016/B978-0-08-092639-1.50043-5>.
- [24] B. McClintock, “The origin and behavior of mutable loci in maize,” *Proceedings of the National Academy of Sciences*, vol. 36, no. 6, pp. 344–355, 1950, doi: 10.1073/pnas.36.6.344.
- [25] J. Duitama *et al.*, “Large-scale analysis of tandem repeat variability in the human genome,” *Nucleic Acids Research*, vol. 42, no. 9, pp. 5728–5741, 2014, doi: 10.1093/nar/gku212.
- [26] A. Blumer, J. Blumer, D. Haussler, A. Ehrenfeucht, M. Chen, and J. Seiferas, “The smallest automation recognizing the subwords of a text,” *Theoretical Computer Science*, vol. 40, pp. 31–55, 1985, doi: [https://doi.org/10.1016/0304-3975\(85\)90157-4](https://doi.org/10.1016/0304-3975(85)90157-4).